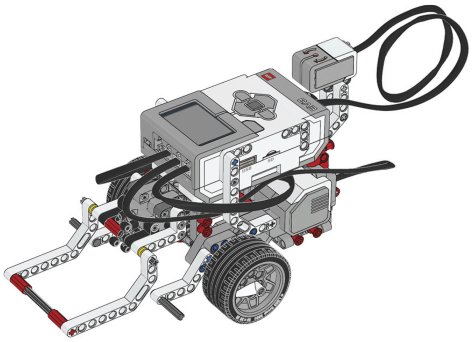
SITE 1101: Principles of Information Systems

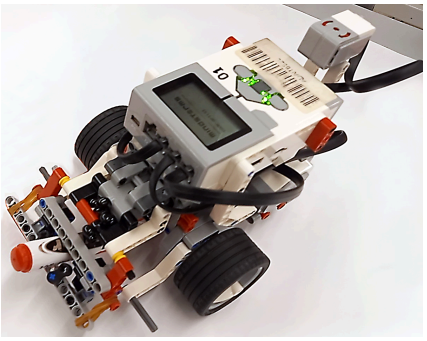
Team Project: Robotics

The true delight is in the finding out rather than in the knowing. ~ Isaac Asimov

Robots help automating many processes, from inventory management[1] to carving marble statues[2]. In this project, you will program a robot prototype that draws lines by moving on a surface, inspired by iRobot® Root® robot[3]. Watch the introduction to the homework: https://youtu.be/E45DRRK683o.

You will use **LEGO® Mindstorms® EV3**, which is the third generation of LEGO robots. By combining LEGO elements with a programmable brick, motors and sensors, you can make your models walk, talk, grab objects and do almost anything you can imagine. Check out https://www.lego.com/en-us/themes/mindstorms/ev3





LEGO artist robot is a base model with a gyro sensor and a medium motor with marker holder.

Preparation

Check time slots that are free in the **B013** (Robotics Lab) in the visits schedule: https://adauniversity-my.sharepoint.com/:x:/g/personal/ishahaliyev_ada_edu_az/IQDwrzfOY1z_QKRqHH1GqxLxAbAYIWUkOuTQ8_R6FND9ygg. **Choose and agree** with your team time slots. Visit the Lab and meet Mr. Nariman Vahabli (nvahabli@ada.edu.az), Lab Coordinator (LC) there to **book those time slots** for your team. There are a limited number of robots, so each team will be able to spend only limited time in the lab, i.e. **4 visits, 1 hour each**. You cannot have more than two visits on one day. Make sure **every team member** participate in **at least 2 lab visits**.

Download and install the software for programming (JavaScript editor) https://makecode.mindstorms.com/offline-app. **Explore the tutorials** that are available in the website and online. You can access Robot Trainer materials also at https://education.lego.com/en-us/lessons/ev3-robot-trainer. You may not be able to connect to EV Brick in tutorials, but make sure you go through all materials to have full understanding.

Start a project with the editor with [New Project] button to write a simple program for your robot model in the software you have downloaded. You can use both Blocks and JavaScript for development (make sure to explore both).

Decide the artwork your robot will emulate by sketching it on paper. Perhaps it will be a famous one everybody would recognize. Number of turning and intersection points on the sketch will be used as a measure of complexity, the more is the better (e.g. octogram has eight intersection points, square has four,

[1] https://youtu.be/UtBa9yVZBJM
[2] https://youtu.be/_VlkMuo2Zcs
[3] https://youtu.be/yC_Tv-GEN-E

circle has none). Drawing complexity will be assessed subjectively. Your drawing does **not** necessarily have to be a geometric shape.

Engineering Culture

Come prepared to labs with a ready version of your program and a clear plan of what each member is going to do. Work in the designated area i.e. at the assigned work desk in the Robo Lab. You are not allowed to take any parts outside of this area. For any related logistical **emergency** email the LC at nvahabli@ada.edu.az immediately, and put your professor(s) on copy (CC).

During the lab visits, if needed, LC will ask you to sign necessary document(s), will note down the inventory ID number for the robot you received, start and finish time of your visit, and will check if the elements are intact. **Note:** You cannot start working before the LC checked you in, and leave the lab before the LC checked you out.

Demonstrate engineering culture of keeping your workplace **tidy and functioning**. After each lab visit you will clean and tidy up your work desk e.g. recycle all used items (e.g. draft paper or plastic bottles), return all the furniture to the previous position.

Development

Follow the instruction manual to build the **Base Model** of Robot Educator with a gyro sensor and medium motor extension attached (see the picture), in your **first lab visit**. In the physical copy of the manual[4] that will be provided to you, pages 7-40 illustrate the steps required for building the robot.

You need to develop and add a custom **marker holder**. You can see the picture to get a hint on of one option on how to build such a holder.[5] Pages 54-67 demonstrate how to add medium motor extension used in this option. Note that, marker holder may require from you to improvise and demonstrate your engineering skills.

Run the sample program on blackboard (`code-artist.uf2`) that moves the robot. Put the robot to the ground before uploading the software.

- Move your motors with lower speed to understand the behavior of the robot
- Keep in mind that "wait for..." blocks do not stop motors, just delay execution of the next block until some condition is met
- Keep in mind that "move (right/left)" blocks do not move the robot accurately
- Use the Gyro sensor for accurate detection of rotation angles of your robot. Pages 48-50 in the instruction manual demonstrate how to connect Gyro sensor.
- When running your program on a robot, it is a good idea to check the cables, as they get loose disconnecting and reconnecting solves the issue of wrong readings.

Write a program to make your autonomous robot to draw the sketch you planned, not bigger than the size of an A1 flipchart paper. Make sure to include *comments* in your code that explain the generation of the sketch.

Submission

Record a video which demonstrates the robot running your final program. After lab sessions, edit your video showing the construction process and the final output by all team members. Add **description** to the

[4] Instruction manual is available at https://robotsquare.com/wp-content/uploads/2013/10/45544_educator2.pdf (pages are slightly different).
[5] Motors are connected to ports B and C, color gyro sensor to ports 2 and 3 and medium touch motor sensor with attached antennae marker holder to port A1.

video with links/credits to the resources or tutorials that were helpful during your development process. One or more team members can narrate the steps and explain correct or incorrect reactions of the robot. Share the video with tags #Mindstorms, #SITE1101, #CodeArt, #BotlarYurdu and #ADAUniversity on Youtube (optionally also on other social media). The most interesting videos will be eligible for **10% bonus**. **Note:** If music is used, use copyright free music so that Youtube won't block the video.[6]

Submit a Microsoft Docx file to blackboard before deadline with the following information:

1. Title (e.g. SITE_1101_Team_Project_K)
2. Team allocation table as shown below. Contribution percentages must be fair and correspond to reality. For well-managed teams, difference is usually maximum of 10%.[7]

Team N	**Contribution (%)**	**What was done?**
Team Member 1 (Team Leader)	25	Detailed description of the work contributed.
Team Member 2	25	Detailed description of the work contributed.
Team Member 3	25	Detailed description of the work contributed.

3. Link to Youtube video(s) (as a full link: i.e. www.youtube.com/…).
4. Copy-Pasted JavaScript code of your minstorms program which will work.
5. Optionally, any other notes you would like to inform about.

The **grading rubric** is as follows[8]:

An engaging video of all contributed team members, demonstrating the construction, development, and the final capacity of the robot, with clear explanations of what worked and what didn't	No video, link doesn't work, or a very bad video.	Explanation by team members exists but no construction / development process is shown.	Construction / development process by contributed team members is there, but the explanation is unclear / we are not sure if the robot works.	Construction / development process and clear explanation is there, but the solution is incomplete.	Construction and development process by contributed team members and a clear, step-by-step explanation of the mechanics of the working robot is demonstrated.	7
Submission format corresponds to the instruction requirement.	Incorrect file format.		One or more required items are missing.		The file format is right, everything is complete, tidy and clear. JavaScript works.	3
Bonus 1: Video quality	Low-quality, boring video.		There was some attempt to make the video engaging.		Video is high-quality and, interesting.	1
Bonus 2: Art Complexity					Robot draws the complex drawing without any issues.	1

When submitting, make sure to **consider video upload time** and other logistical issues. Deadline policy is noted in the syllabus. **Good luck!**

[6] Optionally, upload to Vimeo or Dailymotion as a backup.
[7] If a team member doesn't contribute, their contribution is 0%. You must inform your instructors about it as soon as possible by following the communication rules.
[8] Rubric is supplementary to the requirements noted in the document previously. If the project is fulfilling neither requirement in the closest proximity, it may receive an average value. For example, if your project is neither fully complete (100%) nor worth the 75% criterion, then you may receive 87.5%.